



Propeller Partial Ground Effect

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We extend our recent electric propeller ground-effect study to consider a circular ground-plate, instead of planar-surface of assumed infinite extent. Parameter studies include propeller to plate diameter ratio, propeller diameter to ground-offset ratio, and propeller pitch to diameter ratio. As with classical ground effect, benefits of thrust-augmentation and/or power-reduction with proximity to the ground, depend on the propeller pitch to diameter ratio, with stronger ground-effect for smaller pitch to diameter ratio. Flow visualization suggests on/near the ground-plate suggests a projected dividing-streamline, external to which, the flow is largely radial. A ground-plate of half of the propeller diameter was found to have minimal distinction from that of an unimpeded free-stream, while when the plate and propeller have the same diameter, the resulting ground-effect already resembles that of the infinite-plate.

I. Nomenclature

B	=	Propeller blockage ratio, or ratio of ground-plate diameter to propeller diameter
C_T	=	Propeller thrust coefficient; $C_T = \frac{T}{\rho n^2 D^4}$
C_P	=	Propeller Power Coefficient; $C_P = 2\pi C_Q = \frac{P}{\rho n^3 D^5}$
C_Q	=	Propeller Torque Coefficient; $C_Q = \frac{Q}{\rho n^2 D^5}$
D	=	Propeller Diameter, (m)
d_p	=	Ground-plate Diameter, (m)
D_p	=	Drag on Ground-Plate, (N)
h	=	Distance to the ground-plate, (m)
n	=	Propeller RPS
P	=	Power, (W)
p	=	Pressure, (Pa)
γ	=	Propeller Pitch, (in)
Q	=	Torque (Nm)

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$$\begin{aligned}
R &= \text{Propeller Radius (m)} \\
T &= \text{Thrust, (N)} \\
T_{\text{eff}} &= \text{Effective Thrust, (N)} \\
\rho &= \text{Air Density, (kg/m}^3\text{)} \\
\sigma &= \text{Propeller Solidity; } \sigma = \frac{s}{0.25\pi D^2}
\end{aligned}$$

Subscripts:

$$\begin{aligned}
OGE &= \text{Out of Ground Effect (~1.5 } h/D\text{)} \\
IGE &= \text{In Ground Effect}
\end{aligned}$$

II. Introduction

Ground effect on lifting rotor/propeller performance has been studied theoretically [1] and experimentally ([2], [4]-[6], [8], [11]) for decades. The general idea is that placing the propulsor adjacent and parallel to a ground-plane offers two benefits: an increase in thrust-produced, and a decrease in power-required, for a given rotational speed. Historically, most investigations focus on helicopter rotors, which rarely have a spanwise twist. Here we follow the widespread burgeoning of small-scale propulsor experiments, made possible by ever-improving components from electric radio-control “hobby” aviation, to consider parameter variations not broadly available in the literature. In particular, we consider the extreme ground effect, where the ratio of ground plane stand-off distance to propeller diameter is 0.1 or less. We reverse the propeller, to examine as it were positive and negative ground effect (ceiling effect).

Our previous study ([19]) assumed an infinite ground-plane, where the ratio of ground-plane physical extent to that of the propeller diameter was large. Here, we consider as ground-planes circular plates, of various plate diameter to propeller diameter ratios. This “blockage” type study might give an insight into changes in propeller performance with proximity of fuselage diameter in propeller driven airplanes under pusher and puller configurations ([17]). Studies on propeller-fuselage blockage effect has been done on both large-scale propellers and R/C size propellers at different advanced ratios. Experiment done by Weick ([12], [13]) on a large-scale propeller suggests that the maximum propeller efficiency increases by 1% with a 3% reduction of blockage ratio at range of 0.35 to 0.42. A decrement of 4% in the maximum efficiency was found with a smaller propeller-fuselage distance for a 10’5” propeller. Verstraete and MacNeill’s experimental result for an R/C-sized propeller [14] indicates that with increase in blockage ratio ($B = d_p/D$, where d_p is the plate diameter and D is the propeller diameter), the thrust and power coefficient decrease when compared to the respective coefficients at infinite distance from the plate. Here, we extend this result to other values of blockage ratio, adding propeller pitch to diameter ratio as another parameter.

III. Experimental Setup

Experiments were conducted on a thrust-stand at the University of Dayton. All propellers used in the experiment are from the “thin electric” APC series, selected because of evidence in the literature for their relatively high efficiency and good geometric repeatability across a broad range of diameters [9], [10]. All propellers were driven by an E-Flite Power 60-400kV brushless out-runner electric motor, driven by a PSW 30-108 constant-voltage power supply. A photodiode is used to measure propeller RPM. A photograph is given in Figure 1, and a schematic in Figure 2.

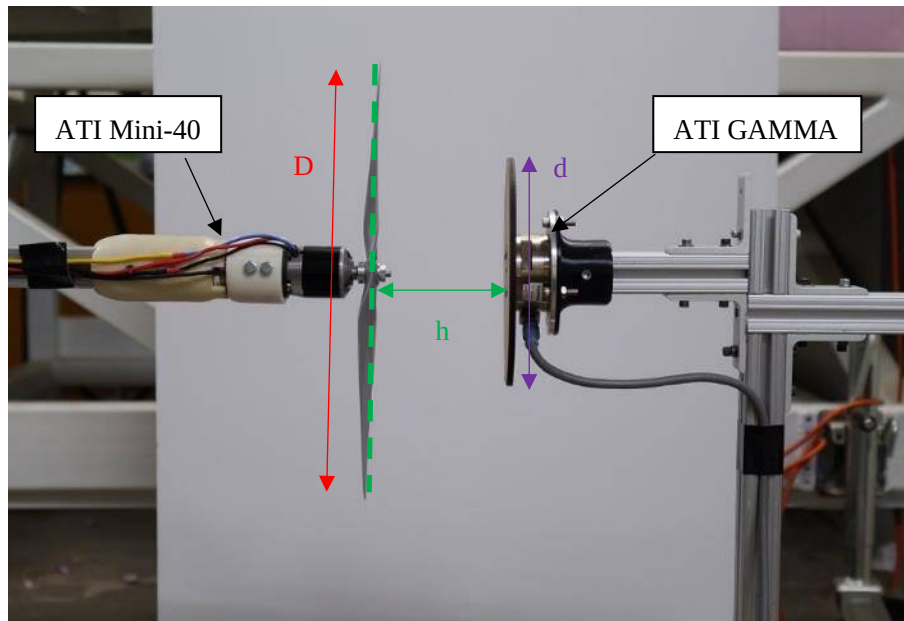


Figure 1. Photograph of thrust-stand (left of image) and circular ground plate.

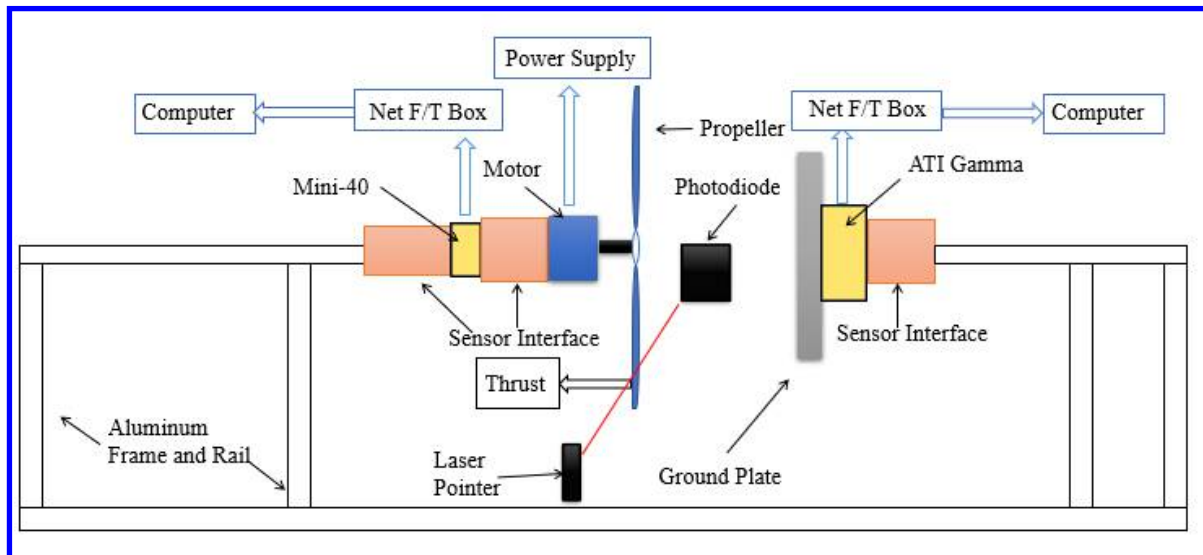


Figure 2. Schematic (side view) of thrust stand and ground-plate

A) Thrust and Torque Measurements

The propeller motor is attached to an ATI Industrial Automation Mini-40 six-component force balance (inside of the plastic shroud shown in the left half of Figure 1; www.ati-ia.com [15]). Thrust is the axial-force along the z-axis of the sensor, and torque is a rotational moment about the z-axis. The balance bolts to an aluminum-extrusion frame secured to the lab floor. The balance has a maximum of 120N force on its Z axis and maximum torque of 2 Nm about its Z axis. The encountered peak thrust and torque were 40N and 1.2 Nm, respectively. The data sampling rate was 100 Hz, for a 15-second duration at each data-point, with two runs to improve data repeatability for each data point (value of rpm; propeller pitch/diameter; and normalized ground-standoff distance h/D). In an effort to minimize sensor-drift, tare values are taken before and after each test, and then the average of the two tares is subtracted from the respective thrust and torque readings.

With the thrust-line oriented horizontally, a vertical wall of $\sim 2\text{m} \times 3\text{m}$ extent acts as a traditional (infinite-extent) ground, and different h/D is achieved by sliding the thrust-stand's frame towards the wall. To attenuate

blockage in the experimental setup, which is to say, blockage unrelated to ground-effect, the propeller-disk is >1 diameter behind any transverse members of the aluminum frame, and the 3D-printed plastic streamline-cowling covering the sensor is $<2\%$ of the propeller disk-area. As is usually the case, power-wires from the speed controller to the motor, cross the metric junction.

To consider partial ground-effect, a circular disk of diameter d_p is mounted a distance h away from the plane of the propeller. The disk is attached to an ATI Industrial Automation Gamma six-component force balance www.ati-ia.com [16], so that the “drag” force on the disk is the axial-force along the sensor’s z-axis. The balance has a maximum of 200N force on its z axis.

To achieve negative h/D , the propeller is attached reversely on the motor. The motor rotation direction is also reversed, by exchanging two of the three wires between the motor and its electronic speed controller. All propellers in the present study have diameter 17”, with variation in pitch (Table 1), to capture effects of pitch to diameter ratio, γ/D . This variation was found in our previous study [19] to be the main factor affecting ground-effect. The blade Reynolds number is calculated at 70% blade-span, for the peak rpm achieved for each given propeller. Thrust coefficients at 6000 RPM are taken and then divided by the solidity of the blade to get C_T/σ .

Table 1. Propeller Properties

Propeller	Pitch (γ)/ Diameter (D)	Solidity (σ)	Peak Tip Speed (m/s)	Peak Reynolds Number ($\times 1000$)	Thrust Coefficient (C_T)/ Solidity (σ)
17x12	0.705	0.0718	124.6	145	1.401
17x10	0.588	0.0753	132.1	160	1.378
17x7	0.412	0.0846	142.0	190	0.889

Following our prior study on infinite ground-plates, $-1.5 h/D$ and $+1.5 h/D$, are taken to be out of ground effect (OGE), as the deviation in measured thrust in ceiling effect and ground effect, at those respective conditions, is small, as compared to thrust-stand testing in free air. Three plate diameters (d_p) were considered: 50% of the propeller diameter, 75% and 100%.

B) Flow Visualization

Tuft flow visualization on the ground plate is performed for all propeller cases listed in Table 1, at RPM ~ 6000 , from 0.1 to 0.5 h/D . The tufts are made of multiple cotton threads attached to a base of size 17”x17”. Each tuft has a length of approximately 1.5”. Smoke visualization was also performed on the 17x7 and 17x12 propellers, at RPM ~ 900 . The smoke wand has a diameter of 0.1”, and was placed behind the propeller. A 200-mW laser was used as a light source for flow visualization.

C) Pressure Transducers

Seven Kulite XTEL-10L-190 pressure transducers (www.kulite.com) [18] were placed flush on the ground plate at normalized radial locations of 0.23, 0.35, 0.41, 0.47, 0.53, 0.58 and 0.7, as shown in Figure 3. All have a range of 170 Kpa. 5 seconds of data was taken for each h/D value at 5,000 Hz sampling rate. Pressure data were taken in the h/D range from 0.1 to 0.5, in 0.1 h/D increments.

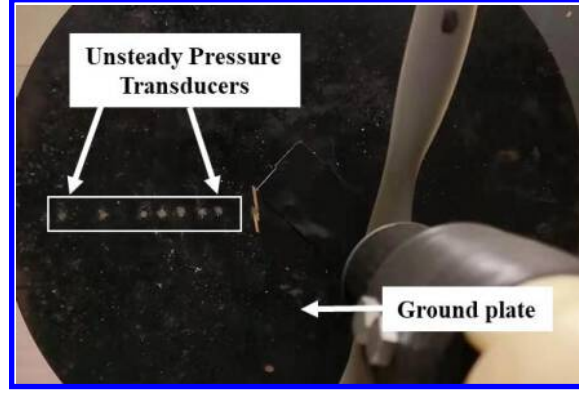


Figure 3. Placement of the Kulite pressure transducers on ground plate

IV. Results

A. Review of Infinite Ground-Plate Thrust and Power Results

A common representation of thrust and torque in ground-effect is “normalized power” (P_{IGE}/P_{OGE}), at constant thrust. To calculate this, a convenient value of rpm is fixed, and OGE thrust and torque (at $h/D = 1.5$ or higher) is calculated by interpolating across the available thrust and torque data points across the rpm range for multiple h/D cases (interpolation is necessary because rpm is not identical from run to run). For every h/D , one finds the in-ground-effect (IGE) rpm at which thrust at that h/D equals the OGE thrust at the reference rpm. At this IGE rpm, one arrives at the power, P . Typical results for P_{IGE}/P_{OGE} are given in Figure 4 for propellers of different pitch to diameter ratios. Each curve in Figure 4 represents the average P_{IGE}/P_{OGE} across the test RPM range. Error bars in Figure 4 indicate two standard deviations in the P_{IGE}/P_{OGE} .

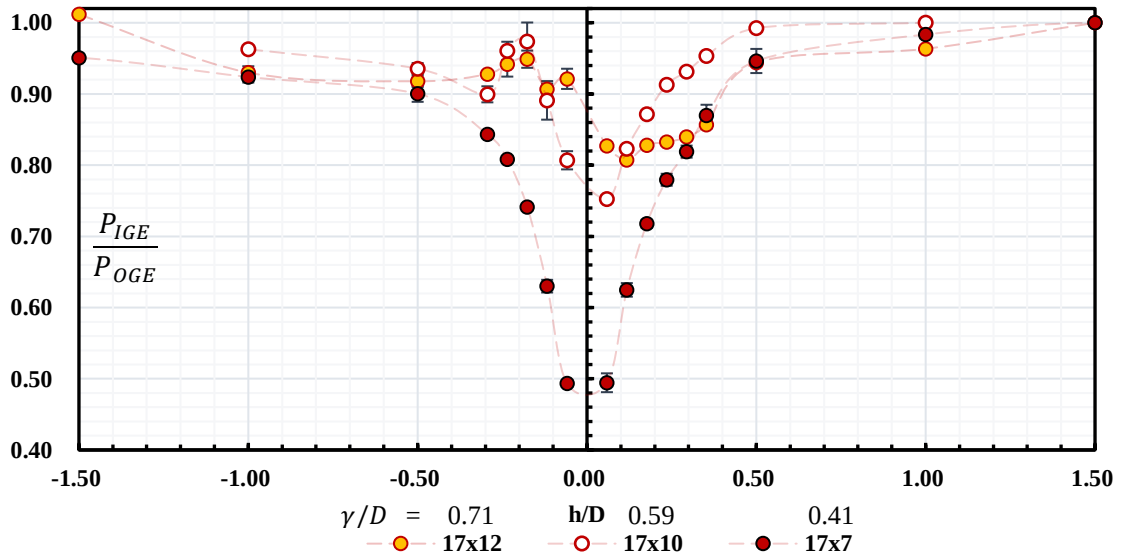


Figure 4. Normalized power required for constant thrust.

For the 17x7 propeller, normalized power at the most extreme ground-standoff, $h/D = 0.05$, is half of the OGE value. The trend of P/P_{OGE} vs. h/D is symmetric, for small positive or negative h/D , but there is a slight

asymmetry between $h/D = +1$ and $h/D = -1$. For the 17x10 propeller, IGE power-reduction is much less dramatic, and is less asymmetric. For $h/D \sim -0.2$, there is an odd reversal in P_{IGE}/P_{OGE} . Still, at least for positive h/D , the variation of P_{IGE}/P_{OGE} with h/D is smooth and orderly, resembling the experimental data in the literature. The 17x12 case is most peculiar, with only a hint of reduction in normalized power in ground effect. The trend is towards larger savings in IGE power, and better positive/negative h/D symmetry, with decreasing ratio of pitch to diameter ratio.

P_{IGE}/P_{OGE} is a succinct presentation-format of IGE results, but it potentially masks to the extent that thrust is increasing, or power-required is decreasing. A reversion to the more direct presentation of thrust and power coefficients is given in Figure . The trends of Figure repeat in both subplots of Figure , with the exception that positive-negative h/D symmetry for the 17x7 case is not as evident. Instead, positive/negative trends in thrust and power combine, as it were, to produce the visually appealing symmetry for the 17x7 case in Figure .

For the 17x7 case, peak thrust-increase appears to be at slightly negative h/D , while peak power-reduction is at commensurately slightly positive h/D . The 17x10 case meanwhile still contains a curious reversal in both thrust and power at $h/D \sim -0.2$. Except the 17x12 case, reduction in power coefficient for the positive h/D range does appear to start near $0.3 h/D$, consistent with Betz's proposed threshold [1].

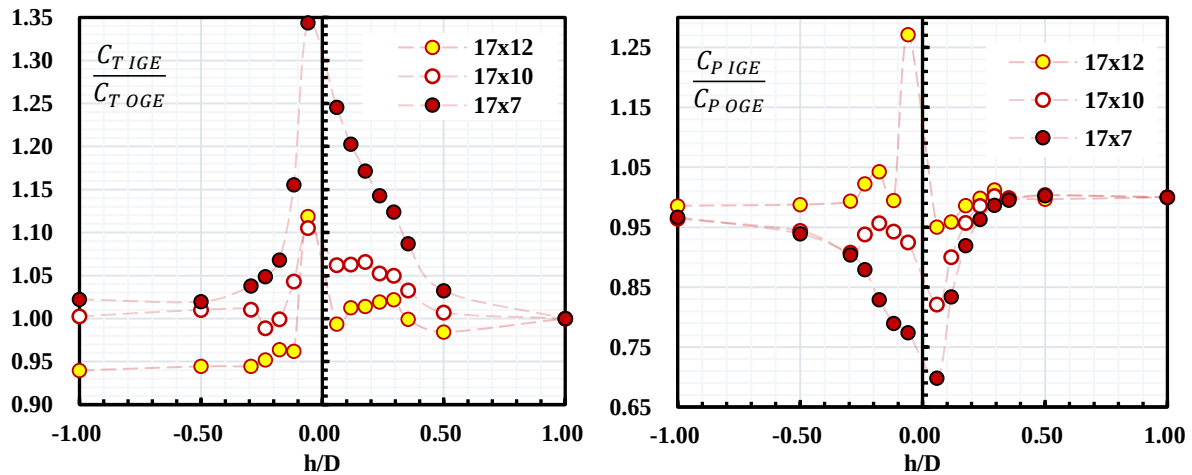


Figure 5. Normalized thrust coefficient (left) and power coefficient (right) for the cases of Figure

B. Thrust and Power in Partial Ground Effect

The power required at constant thrust for plate diameter d_p/D of 50%, 75% and 100% is compared with the case of “infinite” ground-plate in Figure , for the three pitch-to-diameter ratios. Recovery of normalized power to 1.0 at $h/D = 1.5$ implies that this h/D value is effectively OGE. This trend holds for all values of blockage ratio (ratio of plate diameter to propeller diameter), B . Performance of the propeller for $B = 1.0$ closely follows that of the infinite ground plane-case, whether the overall ground-effect trend is strong and obvious (low pitch to diameter ratio) or convoluted (high pitch to diameter ratio). Evidently, $B = 1.0$ is already “infinite”. $B = 0.5$ is the opposite. For the 17x7 propeller, for $B = 0.5$, attenuation of ground effect is some 80%-90% from the infinite-ground case. That is to say, IGE and OGE for $B = 0.5$ bear strong similarity. For the 17x10 propeller, $B = 0.5$ evinces a mild reverse ground effect: normalized power at the smallest h/D exceeds 1. For the 17x12 propeller, the $B = 0.5$ case has a ground effect variation on the order of 50% of that of the infinite-ground, but it should be noted that all cases for 17x12 evince reduced ground effect. The $B = 0.75$ case lies between the $B = 1.0$ and $B = 0.5$ cases, as expected. Around 70% in power required reduction is found for $B = 0.75$ case when compared to power required reduction $B = 1.0$ for all three propellers.

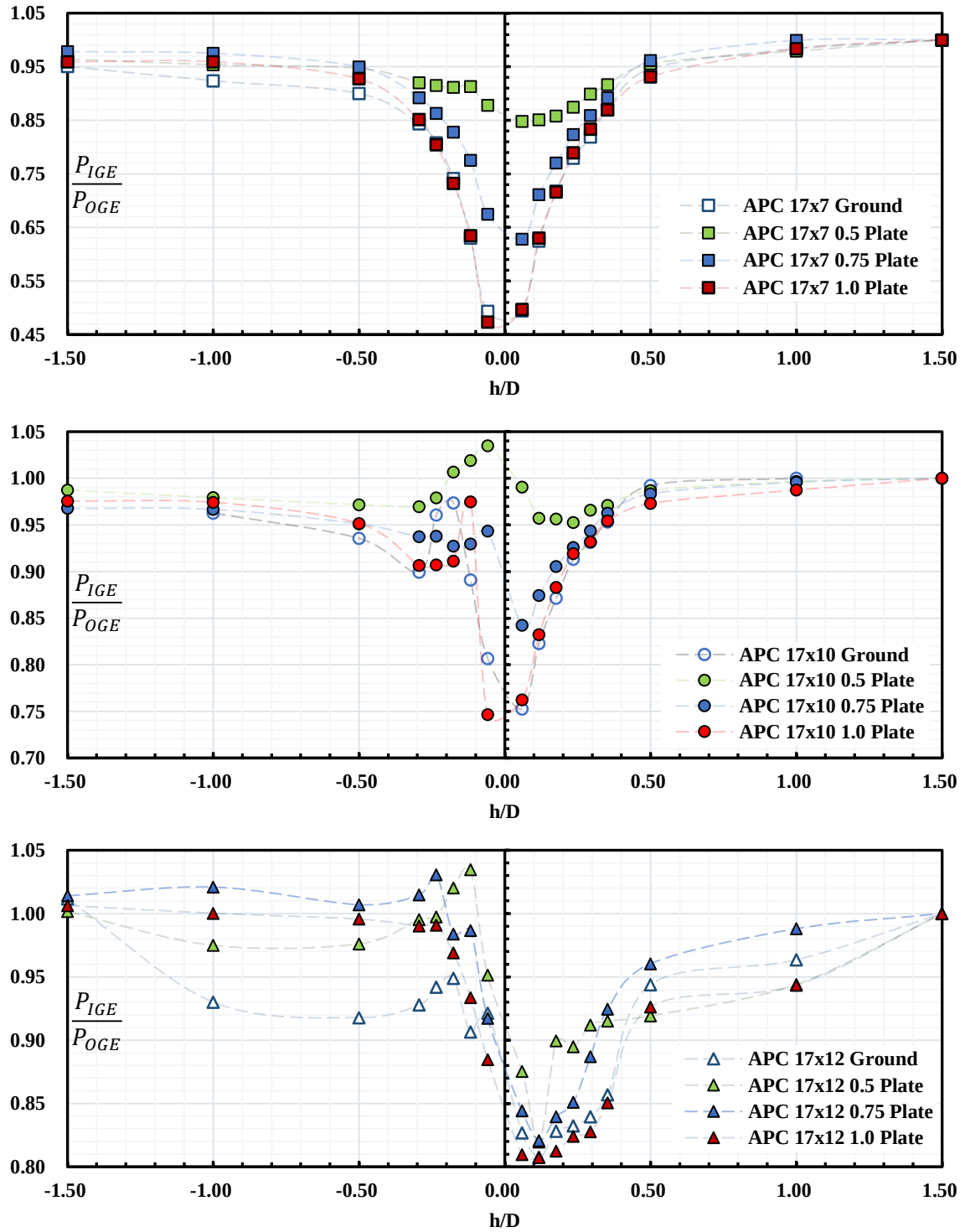


Figure 6. Normalized power required for constant thrust in partial ground effect (17x7,17x10 and 17x12)

The aforementioned trends were examined in more detail in Figure 7 for the 17x7 propeller, which evinces the strongest ground effect, by plotting the individual thrust and power coefficients, normalized by their OGE reference values. For $B = 1.0$, the IGE increment in thrust and decrement in power required overlap the infinite-plate result, which again indicates that $B = 1.0$ can act as a surrogate for “infinite” ground. For $B = 0.5$, changes in power required and thrust-produced are less than 10%. The result for $B = 0.75$ again falls between the 0.5 and 1.0 plate.

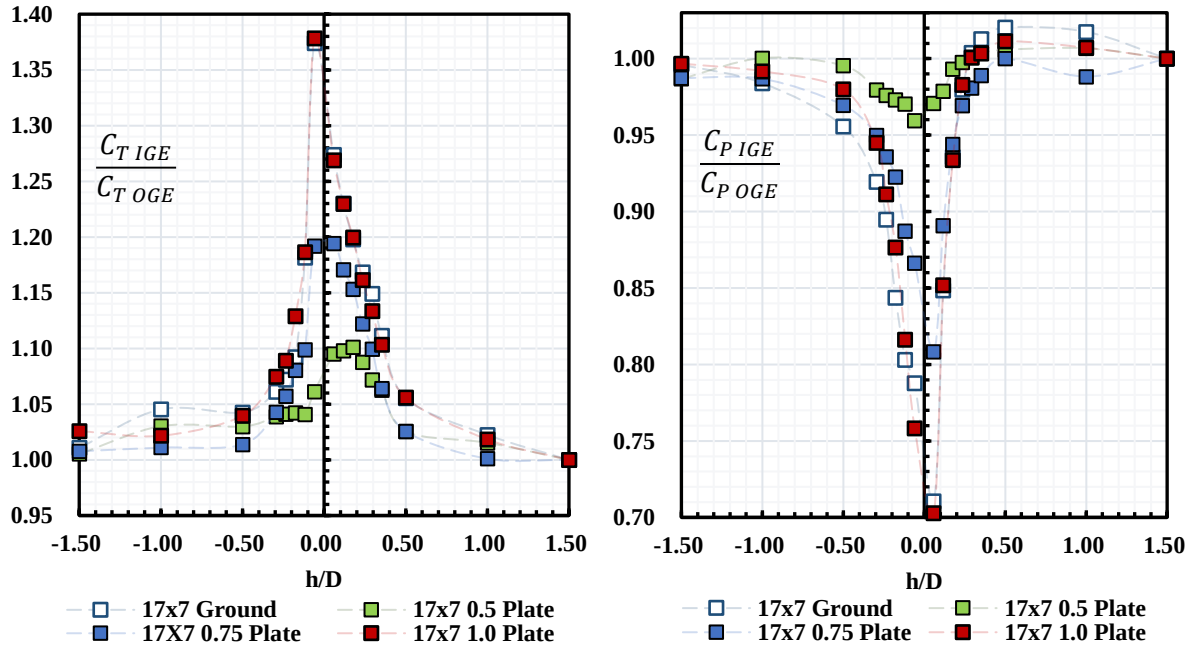


Figure 7. Normalized thrust coefficient (left) and power coefficient (right) for the cases of Figure 6

It is illuminating to consider drag on the ground-plate, for finite plate diameters, due to the propwash. The measured drag force of the plate at each h/D placement is normalized by the OGE thrust, approximated as (T_{OGE} at $-1.5 h/D$), and then scaled by the plate area ratio, B^2 (Figure 8). Using this normalization, the nine curves – combinations of three γ/D and three B values – do not quite collapse to one curve, with some residual variation across B , but very little across γ/D . With the propeller oriented in ceiling effect ($h/D < 0$), drag on the plate relaxes to zero by $h/D = -1$; that is, the propeller no longer has an effect on the plate. The opposite is not true; even for $h/D = 2$, there remains a drag on the plate, as stands to reason, because the propwash theoretically has infinite downstream extent. Positive and negative h/D converge to maximum plate drag at $h/D \sim 0$, which is to say, extreme ceiling-effect and extreme ground-effect result in a plate drag over twice as large as that of far-wake case, $h/D \rightarrow \infty$.

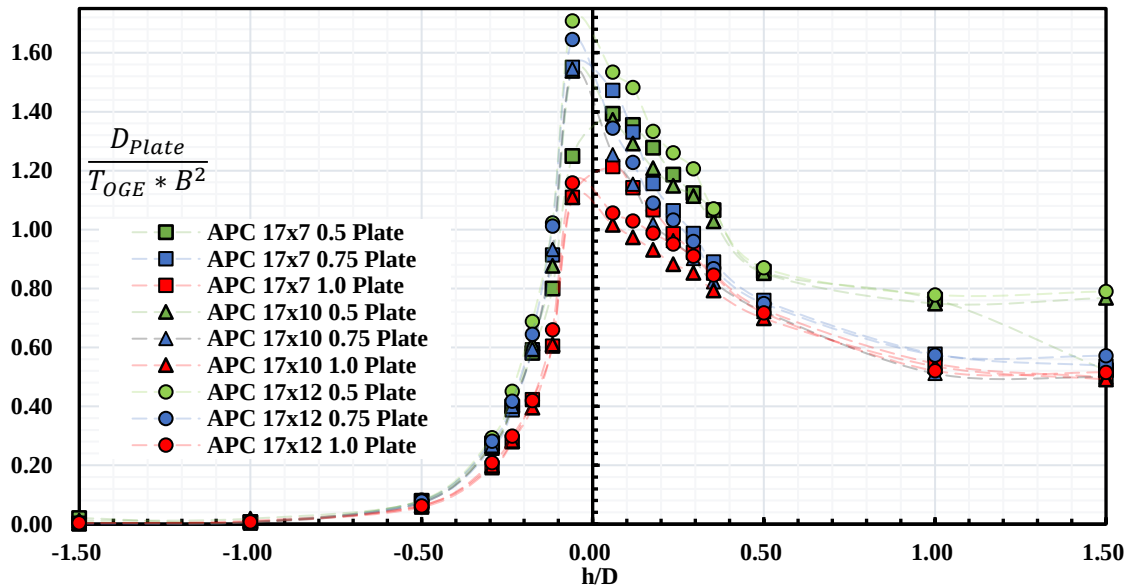


Figure 8. Plate drag normalized by OGE propeller thrust

Using drag on the ground-plate, an alternative presentation of IGE thrust at finite B might be termed as effective thrust T_{eff} : $T_{eff} = T - D_p$ (Figure). This would be the propeller thrust on an airplane in tractor configuration with squared-off fuselage immediately behind the propeller. The effective thrust at each h/D is normalized by approximation of OGE thrust (T_{OGE} at $1.5 h/D$). Unlike the treatment in Figure 8, the nine curves in Figure 9 are not normalized by plate area, so the different B -values (different colors in Figure 9) fall on respectively distinct curves. But as in Figure 8, dependency on γ/D is weak. This means that the strong IGE dependency on γ/D is much attenuated when we include drag on the plate. As in Figure 8, T_{eff} relaxes to the OGE value for the ceiling-effect orientation by $h/D \sim -1$ (pusher configuration where fuselage is in front of the propeller), but in the ground-effect orientation, with positive h/D , there remains residual departure from OGE even at large ground-offset.

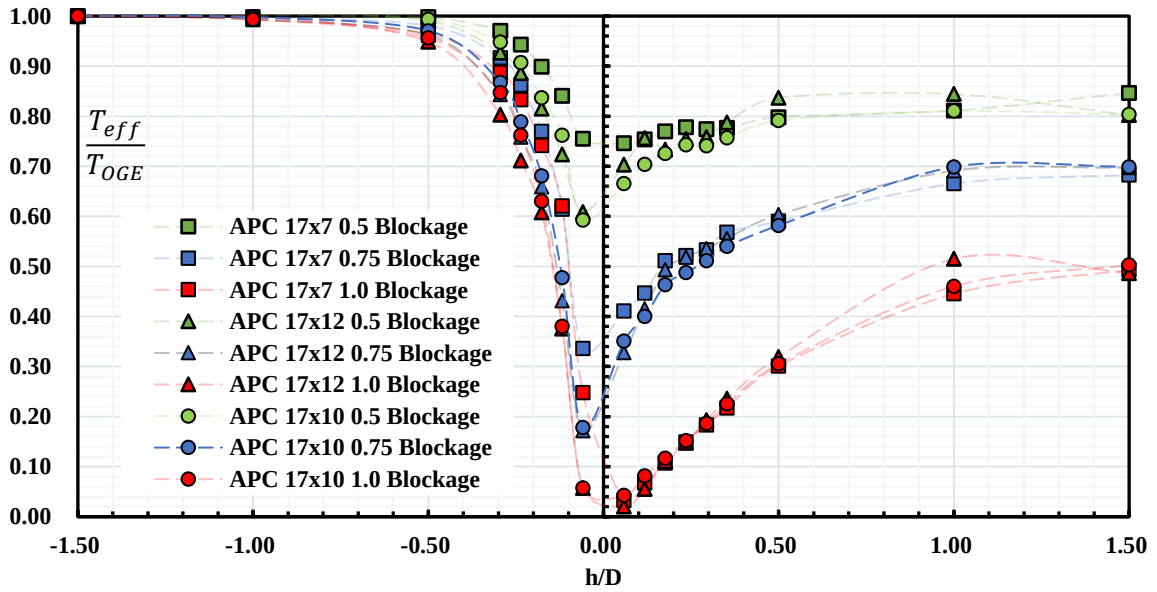


Figure 9. Normalized effective thrust at constant power in partial ground effect

C. Flow Visualization

Tufts on Ground Plates

Tuft-array deflection due to propwash is shown in Figure 10, for high (17x12) and low (17x7) γ/D , for $h/D = 0.3$, for the “infinite” ground plate, and for $B = 1.0$ and 0.5 . Quite prominently for $B = \infty$ and $B = 1.0$, and less so for $B = 0.5$, there is a projected “dividing streamline”, approximately circular, of diameter on the order of half of the propeller diameter. Exterior to this circular boundary, the projected streamlines appear to be radially outward. Interior to the circular boundary, the projected flow can be described as being swirling, without discernible radial component.

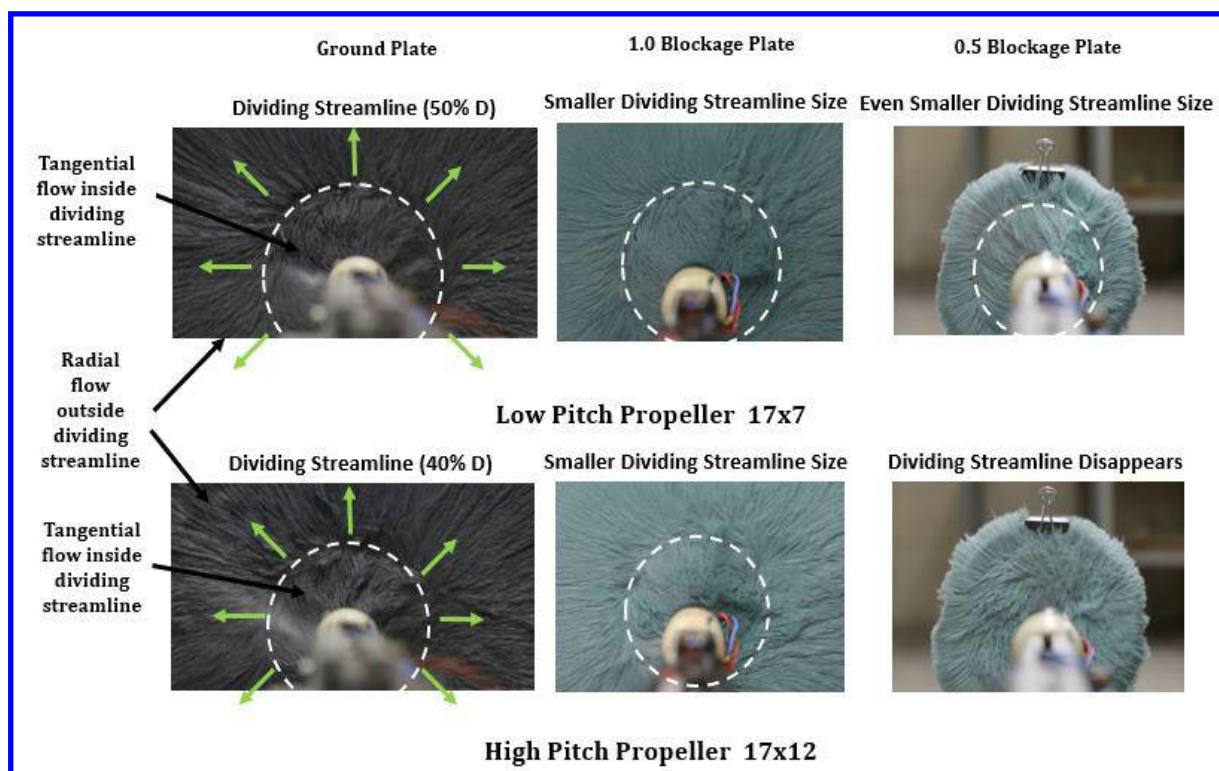


Figure 10. Tuft flow visualization of near-surface flowfield on the ground for high and low $\frac{\gamma}{D}$, $h/D = 0.3$.

Figure 11 plots the diameter of these “dividing streamlines” vs. h/D . Diameter is approximately constant over $h/D \sim 0.1-0.3$ or 0.4 , but for higher h/D , the circular streamline disappears. Such circular dividing streamlines are thus conjectured to correlate with ground effect in some fundamental way. Diameter also decreases slightly with increasing γ/D . For $h/D = 0.1$, $B = \infty$ and $B = 1.0$ cases nearly overlap, but diverge slightly for larger h/D . $B = 0.5$ consistently evinces the smallest dividing-streamline diameter, though not by much – perhaps surprising, given that the streamline comes so close to the $B = 0.5$ plate’s edge.

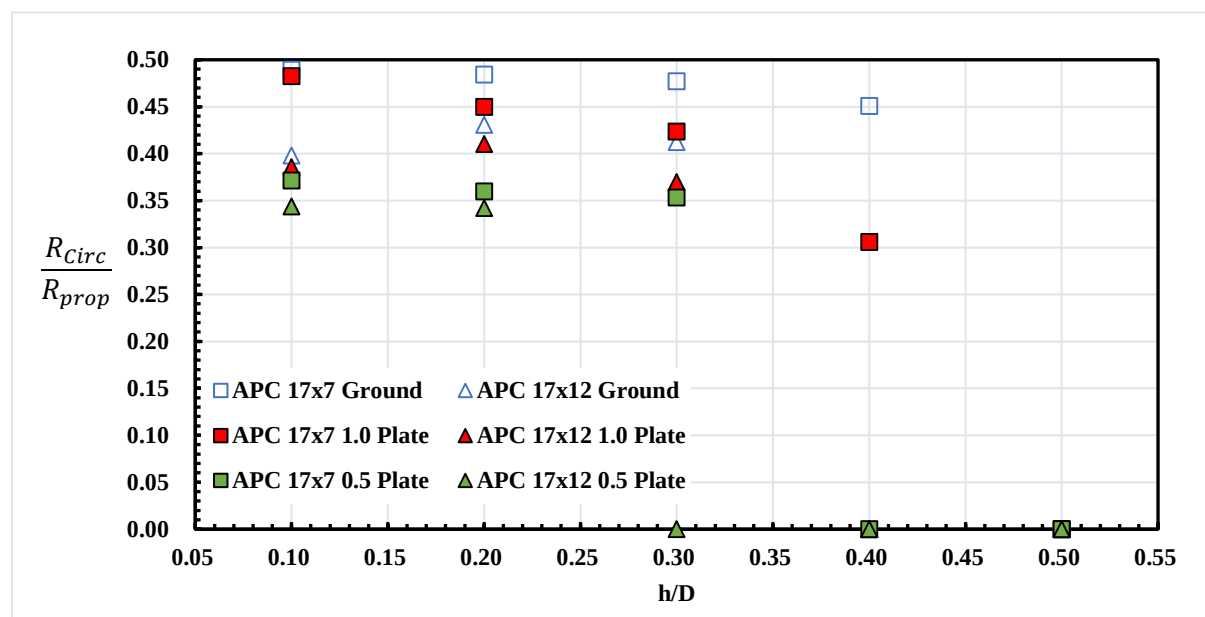


Figure 11. Ratio of Diameter of Dividing Streamline, to Propeller Diameter

Smoke Flow Visualization

Ground-plate tuft results informed placement of the smoke-wand with respect to the projected dividing streamline. Smoke emanating from the wand is concentrated towards the center of the propeller (Figure 12), and stays within the dividing streamline. By moving the smoke wand towards the tip of the blade, the dividing streamline can be observed, where the direction of the smoke switches from center-seeking to radially outboard. For the highest pitch propeller, this deviation happens around 38% of the propeller radius, while for the low pitch propeller case, this happens at around 50% of the propeller radius. In no case is there an obvious entrainment of smoke into propeller tip-vortices, or pile-up of tip vortices, perhaps illustrating a limitation of the technique.

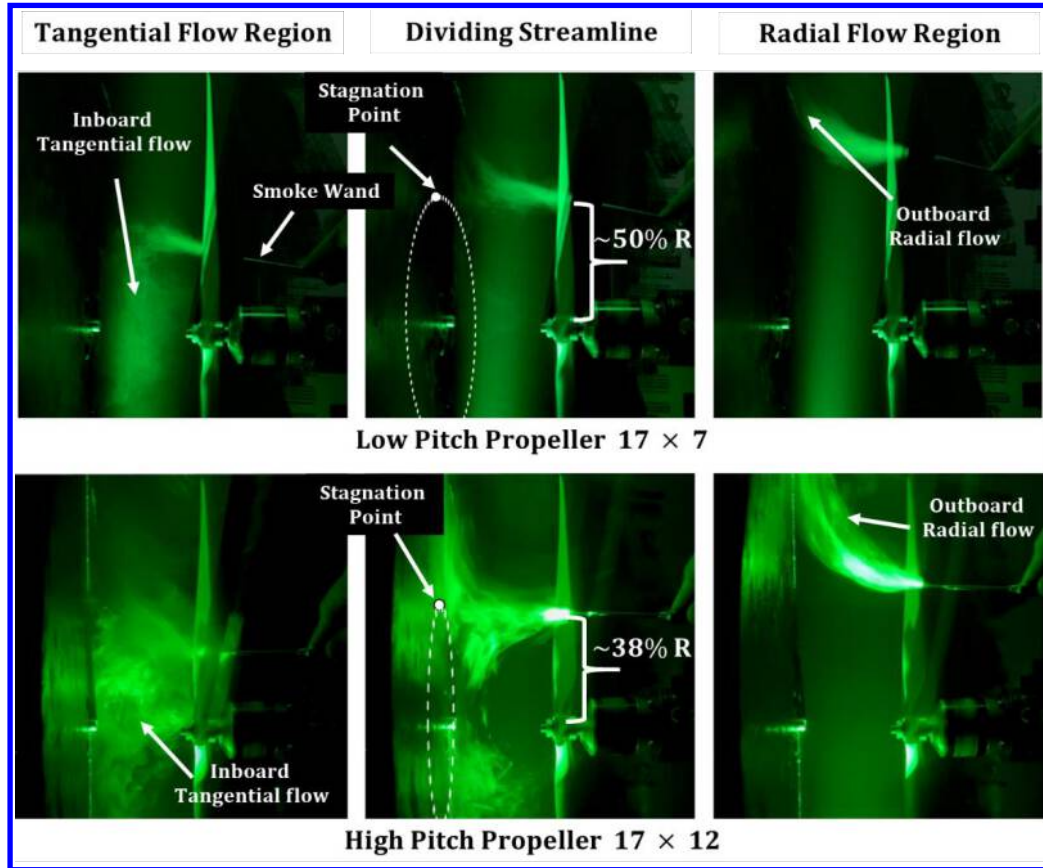


Figure 12. Smoke Flow Visualization on 17x7 (Upper) and 17x12 (Lower) Propellers

D. Pressure Data

Variation of pressure with respect to disk radius is shown in Figure 13 for the 17 x 12 and 17 x 7 propellers for the $B = 1.0$ case. The 17x7 propeller has an RPM of 6,100 and the 17x12 propeller has an RPM of 5,400, such that both produce $\sim 34.5\text{N}$ thrust. The magnitude of the pressure $P(r)$ is normalized by the drag force experienced by the ground plate, and divided by the area of the ground plate. Normalized pressure thus becomes a relative distribution.

For each h/D value in Figure 13, $P(r)$ attains its maximum value in the vicinity of $r/R \sim 0.5$. The actual such value of r/D varies slightly with h/D , with evidence for a larger radius for the lower γ/D propeller. This is consistent with the tuft visualization, in the sense of a dividing projected streamline corresponding to a locus of flow reattachment to the ground plate, where presumably the local static pressure reaches a maximum. It remains

to explain on physical grounds how size of the dividing projected streamline relates to relative IGE thrust-benefit for smaller γ/D propellers.

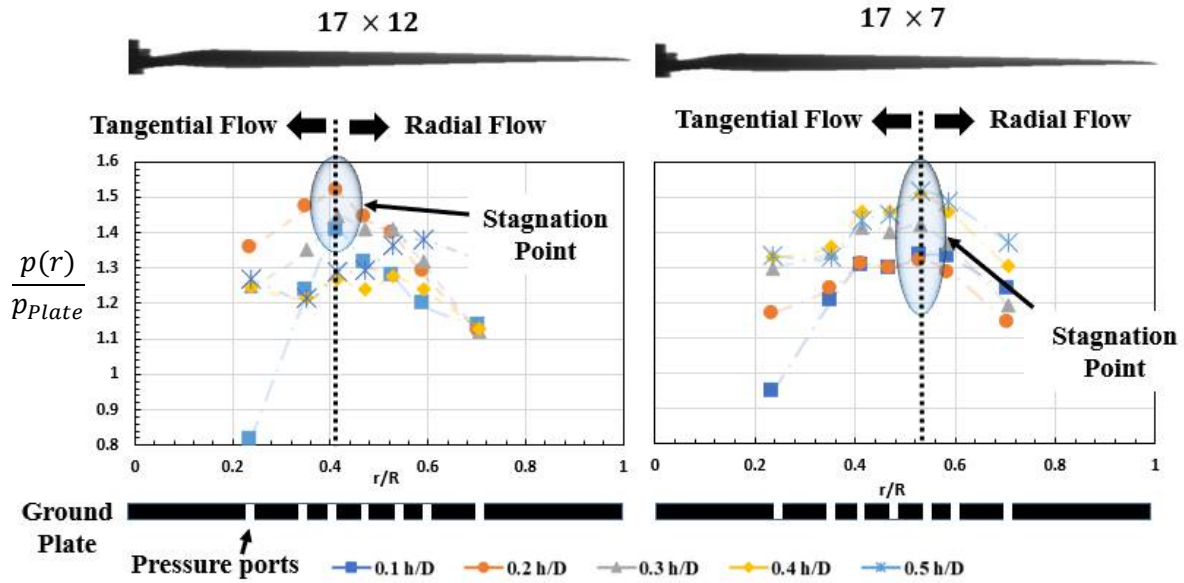


Figure 13. Pressure data vs. propeller radial station, $B = 1.0$. 17x12 propeller (Left) and 17x7 (Right)

V. Conclusion

Finite-diameter ground effect was represented by circular plates, normal to the propeller axis, with ratio of plate diameter to propeller diameter of 0.5, 0.75 and 1.0. As with the case of ground plate of “infinite” extent, the traditional ground-effect-benefits of lower power-required and higher thrust-produced, are greater when the ratio of propeller pitch to diameter is smallest, and lowest – or nonexistent – when this ratio is largest. These effects are symmetric, with in direction of thrust with respect to the ground plate; that is, ground effect or ceiling effect. However, if the force on the plate itself is subtracted from the propeller thrust, then the net result becomes independent of pitch to diameter ratio. For the 0.5 diameter-ratio plate, ground-effect benefits, in thrust-augmentation and reduction in power-absorbed, followed closely those of free-air, suggesting that further reduction in plate diameter largely obviates the ground-effect concept. On the other than, for a plate of 1.0 diameter-ratio, ground-effect benefits followed those of “infinite” ground, suggesting that any further increase in plate diameter is not necessary for a ground-interaction study, at least in terms of force/power measurement at the propeller shaft.

Flow visualization by high-density tuft matrix on the ground-plate shows a circular dividing projected streamline. The lower pitch to diameter ratio propellers have a bigger dividing streamline diameter. This was corroborated by static pressure measurements on the ground plate. How this relates to stronger ground-effect benefits for lower pitch-to-diameter propellers, remains to be elucidated..

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